

Project Estimation

Fundamental Estimation Questions

- How much effort is required to complete an activity?
- How much calendar time is needed to complete an activity?
- What is the total cost of an activity?
- Project estimation and scheduling are interleaved management activities.
- Project Estimation is done on the basis of :
 - • Project size
 - • Project Cost
 - • Effort required
 - • Project duration

Software Cost Components

- Hardware and software costs.
- Travel and training costs.
- Effort costs (the dominant factor in most projects)
 - The salaries of engineers involved in the project;
 - Social and insurance costs.
- Effort costs must take overheads into account
 - Costs of building, heating, lighting.
 - Costs of networking and communications.
 - Costs of shared facilities (e.g library, staff restaurant, etc.).

Costing and Pricing

- Estimates are made to discover the cost, to the developer, of producing a software system.
- There is not a simple relationship between the development cost and the price charged to the customer.

Software Productivity

- A measure of the rate at which individual engineers involved in software development produce software and associated documentation.
- Not quality-oriented although quality assurance is a factor in productivity assessment.
- Essentially, we want to measure useful functionality produced per time unit.

Productivity Measures

- **Size related measures** based on some output from the software process. This may be lines of delivered source code, object code instructions, **etc.**
- **Function-related measures** based on an estimate of the functionality of the delivered software. Function-points are the best known of this type of measure.
- Metrics used for project size estimation
- Lines of codes (LOC)
- Functional points (FPs)

Productivity Comparisons

- The more verbose the programmer, the higher the productivity
 - Measures of productivity based on lines of code suggest that programmers who write verbose code are more productive than programmers who write compact code.
- The lower level the language, the more productive the programmer
 - The same functionality takes more code to implement in a lower-level language than in a high-level language.



SOFTWARE PROJECT ESTIMATION

LOC and FP data are used in two ways during software project estimation:

- as estimation variables to "size" each element of the software.
- as baseline metrics collected from past projects and used in conjunction with estimation variables to develop cost and effort projections.
- LOC This is widely used metric for measuring the size of a software system. It counts the number of lines of code in the software system and is used as a basis for estimating the effort required to develop the software.
- Measured in LOC/pm or LOC/pm⁶

Lines of Code Estimation

LOC

- Count every line of source code except blanks (improve code readability) or comments
- (improves code understandability).
- Count each declaration, actual code containing logic and computations
- Advantages:
 - ✓ Easy to count and calculate
- Disadvantages:
 - ✓ Not a good metric for estimation as its highly dependent on the programming language used.

LOC

- Break projects into features
- Estimate lines of code by analyzing past data. For example, a similar component has been created before and consider its optimistic, most likely and pessimistic values.

$$\text{Expected size} = (S_{\text{optimistic}} + 4 * S_{\text{most likely}} + S_{\text{pessimistic}}) / 6$$

- In upcoming example, the values acquired for the task 3D geometric analysis for past data are :

Optimistic = 4600 LOC, most likely = 6900 LOC, pessimistic = 8600 LOC.

So the Estimated LOC will be calculated using above formula :

$$\text{Expected LOC} = \frac{4600 + (4 * 6900) + 8600}{6} = 6800 \text{ LOC}$$

LOC based estimation example 1

- The mechanical CAD software will accept two- and three-dimensional geometric data from an engineer.
- The engineer will interact and control the CAD system through a user interface that will exhibit characteristics of good human/machine interface design.
- All geometric data and other supporting information will be maintained in a CAD database.
- Design analysis modules will be developed to produce the required output, which will be displayed on a variety of graphics devices.
- The software will be designed to control and interact with peripheral devices that include a mouse, digitizer, laser printer, and plotter.

LOC based estimation example 1

Functions	Estimated LOC
2D geometric analysis	5300
3D geometric analysis	6800
UI and control facilities	2300
Database management	3350
Design analysis modules	8400
Computer graphics display facilities	4950
Peripheral control	2100

LOC based estimation example 1

A “person month” is the metric for expressing the effort (amount of time) personnel devote to a specific project.

- As per historic data, it is deduced that:
 - Organization can write = 620 LOC/pm
 - Payment to a personnel = \$ 8000/month
 - $\text{Cost/LOC} = \frac{8000}{620} = 12.9 \approx \$ 13$
 - Effort required = Total LOC / LOC per person-month
 - Effort required for the project = $\frac{5300+6800+2300+3350+8400+4950+2100}{620} = \frac{33200}{620} \approx 53 \text{ person month}$
 - Project cost to be paid = estimated LOC * Cost/LOC
 - = (5300+6800+2300+3350+8400+4950+2100) * 13
 - = \$ 4,31,600

LOC based estimation example 2

- Login and registration module: 1000 LOC
- Search and booking module: 3000 LOC
- Payment and checkout module: 2000 LOC
- User profile module: 1500 LOC
- Feedback and rating module: 500 LOC
- Use historical data to estimate the effort required to develop the project:
- Let's assume that the historical data suggests an average of 600 LOC per person-month and an average cost of \$8000 per person-month.

LOC based estimation example 2

- Calculate the total number of lines of code:
- Total LOC = $1000 + 3000 + 2000 + 1500 + 500 = 8000$
- Cost per LOC = $\$8000 / 600 = 13.3$
- Effort required = Total LOC / LOC per person-month = $8000 / 600 = 13.3$ person-months.
- Project cost = Estimated LOC * Cost per LOC = $8000 * \$13.3 = \$106,400$

LOC based estimation example 3

- Suppose we have a software project that is estimated to be 50,000 lines of code. The development team consists of 5 developers, each with a salary of \$50 per hour. The project is expected to take 12 months to complete. Assuming 160 hours per month per developer and a productivity rate of 600 lines of code per developer per month.
 - ✓ 1. Calculate the total effort required for the project
 - ✓ 2. Calculate the total cost of each developer per month
 - ✓ 3. Calculate the total cost of the project

Function Point Estimation

Function Points

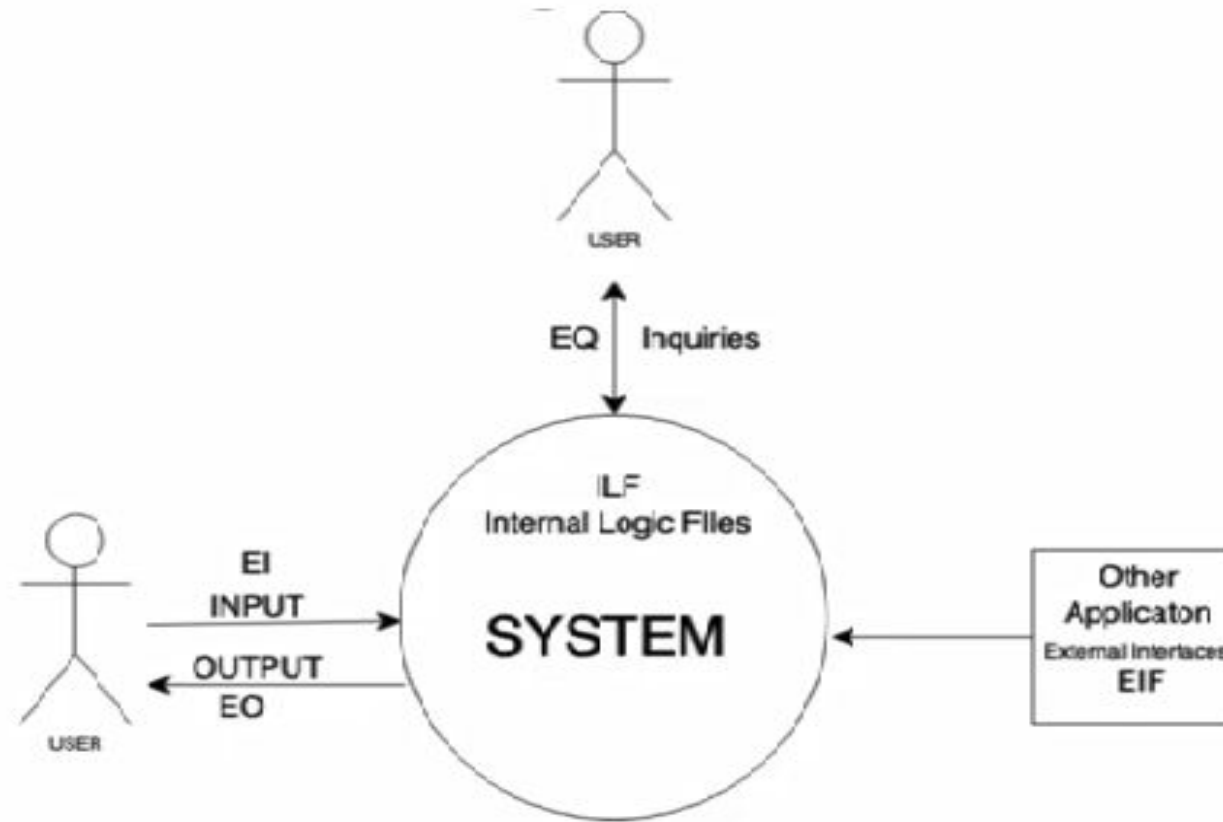
- Based on a combination of program characteristics
 - external inputs and outputs;
 - user interactions;
 - external interfaces;
 - files used by the system.
- A weight is associated with each of these and the function point count is computed by multiplying each raw count by the weight and summing all values.

$$UFC = \sum (\text{number of elements of given type}) \times (\text{weight})$$

$$F.P = UFP \times CAF$$

FP Information Domain

- **Number of external inputs (EIs).** Each external input originates from a user and provides distinct application-oriented data or control information. Inputs are often used to update internal logical files (ILFs). Inputs should be distinguished from inquiries, which are counted separately.
- **Number of external outputs (EOs).** Each external output is derived data within the application that provides information to the user.
- **Number of external inquiries (EQs).** An external inquiry is defined as an online input that results in the generation of some immediate software response in the form of an online output (often retrieved from an ILF).
- **Number of internal logical files (ILFs).** Each internal logical file is a logical grouping of data that resides within the application's boundary.
- **Number of external interface files (EIFs).** Each external interface file is a logical grouping of data that resides external to the application but provides information that may be of use to the application.



1. Computing FPs

Information Domain Value	Count		Weighting factor				
			Simple	Average	Complex		
External Inputs (EIs)		×	3	4	6	=	
External Outputs (EOs)		×	4	5	7	=	
External Inquiries (EQs)		×	3	4	6	=	
Internal Logical Files (ILFs)		×	7	10	15	=	
External Interface Files (EIFs)		×	5	7	10	=	
Count total							

2. Calculating CAF

Complexity Adjustment Factor

$$F.P = UFP \times CAF$$

$$CAF = 0.65 + (0.01 \times \sum Fi)$$

Where,

F_i = Value adjustment factors based on responses to the following 14 questions

2. Calculating CAF

Complexity Adjustment Factor

1. Data Communication
2. Distributed Data Processing
3. Performance
4. Heavily Used Configuration
5. Transaction Role
6. Online Data Entry
7. End-User Efficiency
8. Online Update
9. Complex Processing
10. Reusability
11. Installation Ease
12. Operational Ease
13. Multiple Sites
14. Facilitate Change

Complexity Adjustment Factor is calculated using 14 aspects of processing complexity and these 14 questions answered on a scale of 0 – 5

0 - No Influences or No Important

1 - Incidental

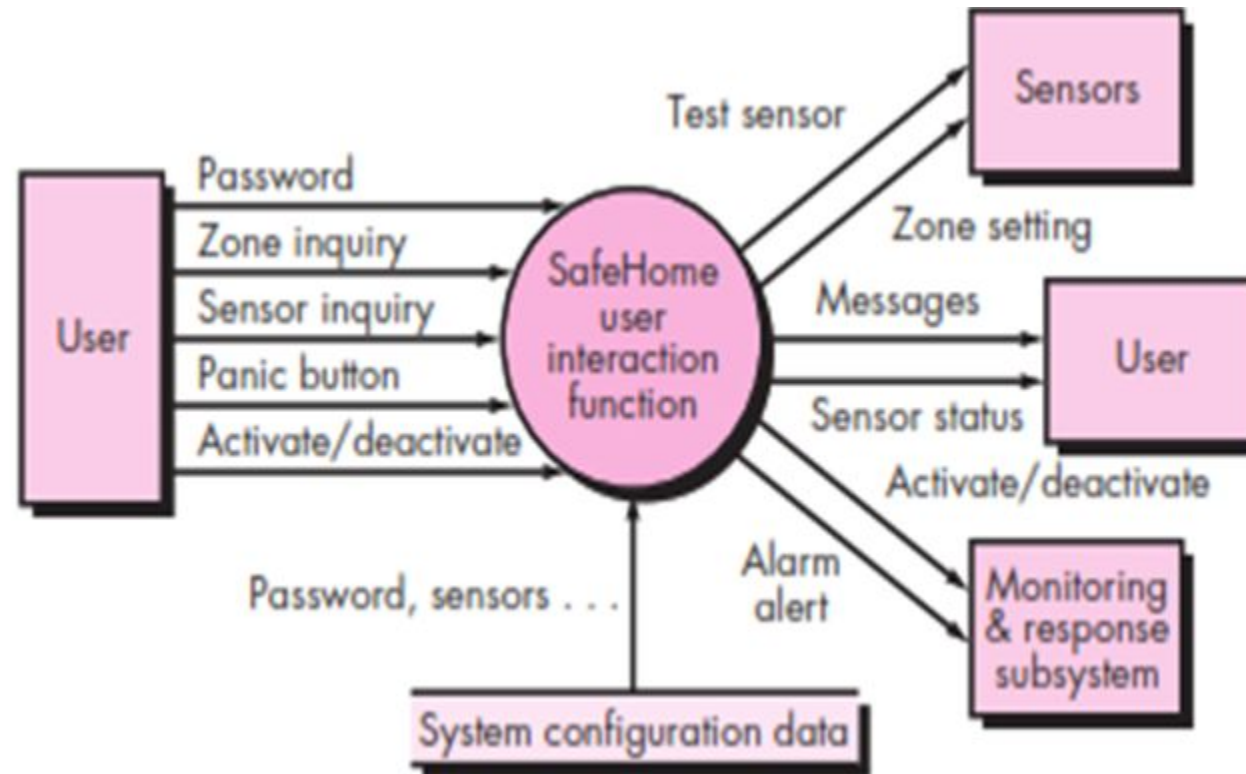
2 - Moderate

3 - Average

4 - Significant

5 - Essential

Example: DFD for computing FP



Computing FP

Information Domain Value	Count		Weighting factor			
			Simple	Average	Complex	
External Inputs (EIs)	3	×	3	4	6	= 9
External Outputs (EOs)	2	×	4	5	7	= 8
External Inquiries (EQs)	2	×	3	4	6	= 6
Internal Logical Files (ILFs)	1	×	7	10	15	= 7
External Interface Files (EIFs)	4	×	5	7	10	= 20
Count total						→ 50

$$F.P = UFP \times CAF$$

$$CAF = 0.65 + (0.01 \times \sum Fi)$$
 Moderately complex product

Then,

$$\sum Fi = 14 \times 2 = 28$$
 2 - Moderate

$$CAF = 0.65 + (0.01 \times 28)$$

$$CAF = 0.93$$

$$F.P = UFP \times CAF$$

$$F.P = 50 \times 0.93 = 46.5$$

Reconciling LOC and FP Metrics /Productivity Estimates

- FPs can be used to estimate LOC depending on the average number of LOC per FP for a given language
 - $LOC = AVC * \text{number of function points}$;
 - AVC is a language-dependent factor varying from 200-300 for assemble language to 2-40 for a 4GL;
 - The relationship between lines of code and function points depends upon the programming language that is used to implement the software and the quality of the design.
 - A number of studies have attempted to relate FP and LOC measures. The following table 4 [QSM02] provides rough estimates of the average number of lines of code required to build one function point in various programming languages:

Programming Language	LOC per Function Point			
	Average	Median	Low	High
Access	35	38	13	47
Ada	154	—	104	205
APL	86	83	20	184
ASP 60	62	—	32	127
Assembler	337	313	91	604
C	162	100	33	704
C++	66	53	20	178
Clipper	38	30	27	70
COBOL	77	77	14	400
Cool Gen/IEF	38	31	10	180
Culprit	51	—	—	—
DBase IV	52	—	—	—
Easytrieve+	33	34	25	41
Excel47	46	—	31	63
Focus	43	42	32	56
FORTRAN	—	—	—	—
FoxPro	32	33	25	35
Idol	66	52	34	203
IEF/Cool Gen	38	31	10	180
Informix	42	31	24	57
Java	63	53	77	—
JavaScript	58	63	42	75
JCL	91	123	26	130
JSP	50	—	—	—
Lotus Notes	21	22	13	25
Mantis	71	27	22	250
Mapper	118	81	16	243
Natural	60	52	22	141
Oracle	30	33	4	217
PeopleSoft	33	32	30	40
Perl	60	—	—	—
PL/I	78	67	22	263
Powerbuilder	32	31	11	105
REXX	67	—	—	—
RPG i3/i4	61	40	24	153
SAS	40	41	33	40
Smalltalk	26	10	10	55
SQL	40	37	7	110
VBScript30	34	27	50	—
Visual Basic	47	42	16	158

FP based estimation Example 2

- User input = 55
- User outputs = 35
- User enquiries = 40
- User files = 8
- External interfaces = 5
- Calculate $UFP = \sum U_i$
- Calculate CAF
- $CAF = 0.65 + (0.01 * \sum F_i)$
- Lets say, the product is significantly complex product so all 14 questions answered in moderately form i.e. 4
- $\sum F_i = ?$ (if complexity factor is differing for each FP then calculate separately.)
- $CAF = 0.65 + (0.01 * ?) = ?$
- $FP = UFP * CAF$
- $FP = ? * ? = ?$

FP based estimation Example 2

Measurement parameter	Counts	Low	Average	High	=count * weighting factor
External inputs	55	3	4	6	?
External outputs	35	4	5	7	?
External enquired	40	3	4	6	?
Internal logic files	8	7	10	15	?
External interface files	5	5	7	10	?
Count total					= ?

Calculate for high complexity

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FP based estimation Example 3

- Suppose we have a software project that has an estimated size of 5000 function points. We have historical data that shows that it takes an average of 10 person hours to complete one function point. The team consists of 8 developers, each with a salary of \$60 per hour. The project is expected to take 12 months to complete. Calculate the effort required and total cost of the project.

FP based estimation Example 3

- To calculate the effort required for this project, we can use the following formula:
- Given that the estimated size of the project is 5000 function points and the productivity factor is 10FP/pm, the effort required would be:
- $\text{Effort} = 5000/10 = 500$ person-months
- To calculate the total cost of the project, we can use the following formula
- $\text{Total Cost} = \text{Effort} \times \text{Monthly Cost per Person}$
- Given that the team consists of 8 developers, each with a salary of \$60 per hour, the monthly cost per person would be:
- $\text{Monthly Cost per Person} = 8 \times 20 \times \$60 = \$9,600$
- Therefore, the total cost of the project would be:
- $\text{Total Cost} = 500 \times \$9,600 = \$4,800,000$

FP based estimation Example 4

Function Type	Low Complexity Weight	Medium Complexity Weight	High Complexity Weight
External Inputs	3	4	6
External Outputs	4	5	7
External Inquiries	3	4	6

Function Type	Number of Functions
External Inputs	10
External Outputs	6
External Inquiries	4

FP based estimation Example 4

- Calculate the Unadjusted Function Points (UFP)
- $UFP = (10 \times 4) + (6 \times 5) + (4 \times 4) = 66$
- Calculate CAF.
- $CAF = 0.65 + (0.01 \times Fi)$ (Assume complexity adjustment factor is average)
- Now, calculate Functional Points

FP based estimation Example 5

- Suppose we have a software project with an estimated size of 50,000 lines of code (LOC) and 5000 function points (FP). We have historical data that shows that it takes an average of 10 person-hours to complete 1000 lines of code, and an average of 10 person-hours to complete one function point. The team consists of 8 developers, each with a salary of \$60 per hour. The project is expected to take 12 months to complete. Calculate the effort required and total cost of the project using both LOC and functional points.

Using Lines of Code (LOC):

- Total person-hours required based on LOC:
- Total LOC = 50,000
- Average person-hours per 1000 lines of code = 10 hours
- Total person-hours = (Total LOC / 1000) * Average person-hours per 1000 LOC
- = (50,000 / 1000) * 10 hours = 500 * 10 hours = 5000 hours
- Total cost of the project based on LOC:
- Total person-hours = 5000 hours
- Number of developers = 8
- Salary per developer per hour = \$60
- Total cost = Total person-hours * Number of developers * Salary per developer per hour
- = 5000 hours * 8 developers * \$60/hour
- = \$2,400,000

Using Function Points (FP):

- Total person-hours required based on FP:
- Total FP = 5000
- Average person-hours per FP = 10 hours
- Total person-hours = Total FP * Average person-hours per FP
- = 5000 FP * 10 hours/FP = 50,000 hours
- Total cost of the project based on FP:
- Total person-hours = 50,000 hours
- Number of developers = 8
- Salary per developer per hour = \$60
- Total cost = Total person-hours * Number of developers * Salary per developer per hour
- = 50,000 hours * 8 developers * \$60/hour
- = \$24,000,000